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Course Code: CS1804 **Name of the Programme: B.Tech (H)** **Name of the Course: Exploring Science** **Semester: 1**		
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	2+0+2	30 + 30 = 60
Course Objectives:		
a) Understand the fundamentals of semiconductor physics, including band theory, intrinsic and extrinsic semiconductors, and the operation of devices like diodes, transistors, and optoelectronic components.		
b) Analyze crystal structures through Bravais lattices, Miller indices, and X-ray diffraction to determine material properties.		
c) Explore the synthesis, unique properties, and technological applications of nanomaterials, along with the basics of quantum mechanics and an introduction to quantum computing concepts.		
d) Examine environmental issues including pollutants, e-waste management, and sustainable methods for water and waste management.		

Course outline (Syllabus of the course)

Syllabus:

Module - 1 **Introduction to Semiconductor Physics and Device Technology**	**No. of hours** **7**
Classification of solids based on band theory · Intrinsic and Extrinsic semiconductors (Concept of electron and holes, mobility) · Diode and its characteristics · Zener diode, Transistor and its input and output characteristics · Photodiode, light emitting diode and solar cells.	

Module - 2 **Crystal Structure**	**No. of hours** **5**
Properties of Crystals · Crystal Lattice (Primitive Cell and Unit cell - Lattice parameters) · Crystal systems · Bravais lattice (Primitive - Body Centered - Face Centered - Base Centered) · Miller Indices · Atomic Packing Fraction (SC - BCC - FCC) · Bragg's law · X-ray Diffraction.	

Module - 3 Nanomaterials	No. of hours 3
What are Nanomaterials ・ Advantages of miniaturization ・ Surface Area to Volume ratio ・ Properties of nanomaterials ・ Techniques of synthesis of nanomaterials (Top down - bottom up - Photolithography - Ball milling - Chemical Vapour Deposition) ・ Nanoscience vs Nanotechnology, Applications of nanotechnology.	

Module - 4 Introduction to Quantum Mechanics and Quantum Computing	No. of hours 10
Pre-Quantum era experiments (Young's double slit experiment - Blackbody radiation - The Hertz Experiment - Planck's Law - Heisenberg's Microscope - Einstein's Photoelectric Equation) ・ Heisenberg's Microscope ・ Principle of Superposition ・ Measurement ・ Observables, Hermitian Operators, Quantum States ・ Schrodinger equation ・ Exactly solvable problems (Free particle - Infinite potential well) ・ A Brief Introduction to Quantum Computing (Qubits, Quantum Gates, Entanglement and Superposition).	

Module - 5 Earth and Environmental science	No. of hours 5
Climate science and modeling ・ Environmental Pollutants (sulphites - carbides - heavy metals - non-chemical - suspended particles) ・ Water Management ・ Waste management (Sources of e-waste - composition - characteristics - Health hazards due to exposure to e-waste - methods of recycling.	

Course outcomes
a) Explain semiconductor behavior, diode characteristics, and transistor operation based on band theory and charge carrier dynamics.
b) Analyze crystal structures using Miller indices and Bragg's law, and interpret X-ray diffraction patterns to determine material properties.
c) Analyze crystal structures using Miller indices and Bragg's law, and interpret X-ray diffraction patterns to determine material properties.
d) Evaluate environmental pollutants, e-waste hazards, and strategies for sustainable water and waste management.

Text Books	
1	Curated lecture notes prepared by the science team of the School of Computer Science and Engineering will be given to the students, as the course is very diverse in nature.

Reference books	
1	***Semiconductor Physics and Devices.*** Author: **Donald A. Neamen** Publisher: McGraw Hill Year of Publishing: 2017 ISBN-10: 0071070102 · ISBN-13: 978-0071070102
2	***Elements of X-Ray Diffraction***. Author: **B D Cullity** Publisher: Pearson Education India Year of Publishing: 2014 ISBN-10: 9332535169 · ISBN-13: 978-9332535169
3	***An Introduction to Nanoscience and Nanotechnology***. Author: **Alain Nouailhat** Publisher: Wiley Publication Year of Publishing: 2008 ISBN-10: 1848210073 · ISBN-13: 978-1-848-21007-3
4	***Quantum Mechanics: Formalism, Methodologies, and Applications.*** Author: **P C Deshmukh** Publisher: Cambridge University Press Year of Publishing: 2023 ISBN-10: 1316512258 · ISBN-13: 978-1316512258
5	***Concepts of Modern Physics.*** Author: **Arthur Beiser** Publisher: McGraw Hill Year of Publishing: 1981 ISBN-10: 0072448481 · ISBN-13: 978-0072448481
6	***Reproducibility and replicability in Science.*** Author: **Arthur Beiser** Publisher: National Academies of Sciences

	Year of Publishing: 2019 ISBN-10: 0309486163 · ISBN-13: 978-0309486163

Lab Programs / Tutorials (Using dedicated kits and SEELab 3.0 Modules)	
1	Study of Error Analysis and Propagation using Vernier Calipers and Screw Gauge.
2	Study the principles of interference using Young's Double Slit Experiment.
3	Measurement of the grating constant and wavelength of light using a Diffraction Grating.
4	Input and output characteristics of a Bipolar Junction Transistor using SEELab 3.0 module.
5	Investigation of IV Characteristics and Responsivity of a Photodiode using SEELab 3.0 module.
6	Verification of Beer-Lambert's Law Through Spectrophotometric Analysis using SEELab 3.0 module.
7	Determination of the Band Gap Energy of a Semiconductor Material using SEELab 3.0 Module.
8	Measurement of dielectric constant using a RC circuit.
9	Investigation of Zener Diode for Voltage Regulation Properties using SEELab 3.0 module.
10	Measurement of Thickness of the Given Material using Air Wedge.